# Junior Deogracias MAMBO MOTSOU

## FPGA & SoC Engineer — VHDL, Vivado, Vitis

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## Profile

Embedded systems engineer (Polytech Dijon, 2026) focused on digital design for FPGAs and Zynq SoCs: RTL in VHDL, AXI4-Lite and AXI4-Stream interfaces, Vivado IP integration and block design, and the bare-metal software that drives the hardware from Vitis. I verify RTL against bit-exact C models with self-checking testbenches. A year of automotive firmware bring-up at FORVIA gives me the system view.

## Technical skills

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| **HDL & design** | VHDL (synthesisable RTL, VHDL-2008 testbenches), Verilog (basic); FSMs, pipelined datapaths, line buffers, bit-growth management, AXI handshakes with back-pressure, bidirectional I/O |
| **AMD/Xilinx flow** | Vivado: IP integrator block design, IP packaging, synthesis, implementation, XDC, timing and utilisation reports, bitstream, ILA and Hardware Manager; Vitis HLS; Vitis bare-metal applications; TCL scripting of the flow |
| **Devices & boards** | Zynq-7000 (ZedBoard, xc7z020), Artix-7 (Nexys 4 DDR, xc7a100t) |
| **Interfaces & IP** | AXI4-Lite, AXI4-Stream, AXI DMA, Zynq PS7 (GP/HP ports), Video Timing Controller, Clocking Wizard, FIFO Generator; VGA, I²C, UART, SPI, ZBT SRAM |
| **Verification** | Self-checking testbenches, bit-exact C reference models, random stall injection; GHDL, XSim, ModelSim; CI simulation on GitHub Actions |
| **Embedded software** | C (bare-metal, Xilinx standalone drivers, CMSIS), FreeRTOS, Zephyr; Python, TCL |
| **Debug** | ILA, JTAG / Hardware Manager, J-Link, GDB, serial consoles |

## FPGA & SoC projects

### Personal

Personal projects. I set the architecture and verification plan; implementation is assisted by AI coding tools. Verified in simulation and emulation, not yet on physical boards.

**Image-filter accelerator for Zynq-7000 with AXI DMA** personal — Oct. 2026
github.com/teo1747/zynq-image-filter-accelerator

- Streaming 3×3 filter IP in VHDL (box, Gaussian, Sobel), one pixel per clock: line buffers, 4-stage pipeline, full AXI4-Stream back-pressure, TLAST/TUSER framing; AXI4-Lite control and status registers; packaged for Vivado IP integrator.
- Zynq PS7 + AXI DMA block design scripted in TCL through bitstream and XSA; bare-metal Vitis application streams frames from DDR, checks every pixel against the C model and times HW against SW.
- Bit-exact against a C reference in a self-checking GHDL testbench with random stalls on both streams (14 frames, run in CI). Board run on ZedBoard pending.

Environment: VHDL, C, TCL, Vivado, Vitis, GHDL, Zynq-7000 (xc7z020)

**EmbSIM — microcontroller and circuit simulator** personal — Oct. 2026 –
github.com/EmbLinkOs/EmbSIM

- Instruction-set simulators and peripheral models (Cortex-M, AVR; STM32F1/F4 GPIO, timers, USART, SPI, I²C, ADC, DMA) coupled to a nodal-analysis circuit solver; CI runs unmodified STM32Cube, Arduino and Pico SDK examples. Early stage.

**EmbCC — C cross-compiler toolchain** personal — Jul. 2026 –
github.com/EmbLinkOs/EmbCC

- C99 compiler, assembler and linker with Cortex-M, RISC-V, AVR and x86-64 back-ends; firmware utilities (SVD-to-header/startup generation, HEX/UF2 packing).
- Verification: encodings cross-checked against llvm-mc/GNU as, generated code run under QEMU and compared with GCC/Clang; GitHub Actions CI on Linux and macOS with GCC and Clang.

### Polytech Dijon

**Real-time VGA video pipeline on ZedBoard** Polytech, pair — 2025 – 2026
github.com/teo1747/zedboard-vga-video-pipeline

- 640 × 480 chain built in IP integrator: Clocking Wizard, Video Timing Controller, AXI4-Stream video IPs, a VHDL inverter and Vitis HLS overlay IPs. Sync signals verified on hardware with ILA; the final demo showed no image because the HLS IPs had no AXI-Lite master to start them, since reworked as free-running (board retest pending).

Environment: Zynq-7000 (xc7z020), Vivado/Vitis HLS 2022.2, ILA, XDC constraints

**ZBT SRAM controller in VHDL** Polytech SoC course, pair — 2025 – 2026
github.com/teo1747/vhdl-zbt-sram-controller

- FSM controller for a Micron 512K × 36 ZBT SRAM: single and 4-word burst read/write on a 36-bit bidirectional bus (Xilinx IOBUF); self-checking testbench against the vendor memory model.

Environment: VHDL, Vivado, Zynq-7000 target, Micron simulation model

**3×3 image filter on FPGA** Polytech, solo — Nov. 2024 – Jan. 2025
github.com/teo1747/vhdl-2d-image-filter

- Streaming averaging and Gaussian (1-2-1/2-4-2/1-2-1) filters on 8-bit pixels: line buffers on the FIFO Generator IP, pipelined adder tree with explicit bit growth (9, 10, then 11 bits); simulated end-to-end on the Lena image, outputs compared with a software reference.

Environment: VHDL, Vivado 2024.1, Artix-7 (xc7a100t), XSim, MATLAB

**FPGA temperature monitor** Polytech, pair — 2024 – 2025
github.com/teo1747/nexys4ddr-temperature-monitor

- Integrated I²C (400 kHz) and UART (115 200 baud) cores to read the on-board ADT7420 and display it on the 7-segment display; timing closed at 100 MHz (WNS +5.4 ns, 1 936 LUTs) on Nexys 4 DDR.

## Professional experience

**Embedded Systems Engineer — apprentice (contrat de professionnalisation)** — Sep. 2025 – Sep. 2026
FORVIA – Faurecia Interior Systems Surface Activation department, France

- Firmware bring-up on the Infineon TLD4020-3ET (Cortex-M23, LIN) for a heating-element controller: LIN commands, PWM via the LDRV peripheral, NTC regulation; root-caused a HardFault to a silicon-revision mismatch, and synchronised ADC sampling to the PWM on-time (mid-pulse ISR, 16-sample average).
- Ported a MeLiBu LED driver on RP2040 (PIO state machines) from the MLX81116 to the MLX81143 for a 154-LED panel; fixed bus timing (break generation, critical-section scope, TX-FIFO drain).
- Owned the software of an interior-lighting demonstrator: multithreaded Python driving nine STM32 UART gateways at 921 600 baud in lockstep, Android tablet control over Bluetooth.

Environment: IAR EWARM, J-Link, Pico SDK, Python/PyQt5, Kotlin, STM32, LIN, MeLiBu, UART

**Embedded Systems Intern** — Apr. – Aug. 2025
KMUTT — MDT Lab Bangkok, Thailand

- Image-processing work for a precision-agriculture project: fruit classification by transfer learning on VGG16 (Keras/TensorFlow), 98.4 % accuracy on a 4 599-image test set (Fruits-360, 131 classes), plus an OpenCV contour/colour detection pipeline.
- FPGA ramp-up: designed an 8-bit microcontroller in VHDL (12-state fetch–decode–execute controller, ALU with NZVC flags, memory-mapped ROM/RAM/I/O) following the LaMeres reference architecture; simulated in Vivado 2024.2 / XSim.

Environment: Python, TensorFlow/Keras, OpenCV, VHDL, Vivado 2024.2

## Education & languages

**Engineering degree (Diplôme d'ingénieur, Master's level) — Embedded Systems** — 2023 – Sep. 2026
Polytech Dijon Dijon, France
Graduated. Final-year apprenticeship at FORVIA. Coursework: digital design, SoC design on Zynq (Vivado, Vitis, Vitis HLS), real-time systems.

**Two-year preparatory programme — mathematics, physics, electronics** — 2021 – 2023
IUC Cameroon

**Languages** English: native (anglophone), TOEIC C1 (2024) | French: fluent, professional use (B2)
**Certification** Cisco CCNA 1 (July 2024)